Accommodation Allocation Engine

Optimal hostel-room assignment for the cultural festival

Part A: Problem Formulation

Track 1 · Mathematical Models for Operations

Proposed approach: Hungarian (Kuhn–Munkres) algorithm + bipartite matching

Group 7 · OOP Summer 2026 · BITS Pilani

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1 · The problem

The festival serves **5,000–6,000 users**. Many are **outstation participants** who need a hostel bed for a few nights, and rooms are a **scarce, constrained resource**: each has a fixed capacity, a gender policy, a price, and may or may not be accessibility-equipped.

Today the warden allocates rooms **by hand on a spreadsheet**. At this scale that causes:

- **Gender / accessibility mistakes**: someone placed in an incompatible room.
- **Ignored preferences**: building, room type, roommates.
- **Wasted beds** sitting empty while other rooms are oversubscribed.
- An **ad-hoc, unfair waitlist** when demand exceeds supply.

> **The need:** an automated tool that computes a room assignment which **minimises total dissatisfaction**, **never violates a hard constraint**, and produces a **fair waitlist**.

2 · Our idea in one slide

We will **model accommodation as the classic *assignment problem*** and solve it with an exact algorithm rather than manual heuristics.

- Treat each **bed** as a slot and each **participant** as something to assign to a slot.
- Score every (participant, room) pairing: **hard rules** (gender, accessibility) are *forbidden*; **soft preferences** (budget, building, room type, roommates) add a *penalty*.
- Find the assignment with the **minimum total penalty** using the **Hungarian algorithm**: the textbook exact, optimal method for assignment.
- Whoever can't be seated goes onto a **priority-ordered waitlist**.

This turns a chaotic manual chore into a reproducible, provably-optimal computation that plugs into the existing platform.

3 · Inputs & outputs

Planned inputs (exported by the Accommodation module as CSV / JSON):

- **Participants**: id, gender, budget/night, nights, arrival day, accessibility need, **category** (Performer / VIP / Delegate / Attendee), preferences (building, room type, roommates).
- **Rooms**: id, building, floor, **capacity**, **gender policy**, price/night, accessible?, type.

Planned outputs (re-imported by the platform):

- `allocations`: who stays in which room, plus the **wallet charge** = price × nights.
- `waitlist`: everyone unplaced, ordered by priority.
- `metrics`: placement rate, dissatisfaction, utilisation.

CSV/JSON keeps the tool aligned with the platform's file-exchange and REST payloads.

4 · Proposed mathematical model

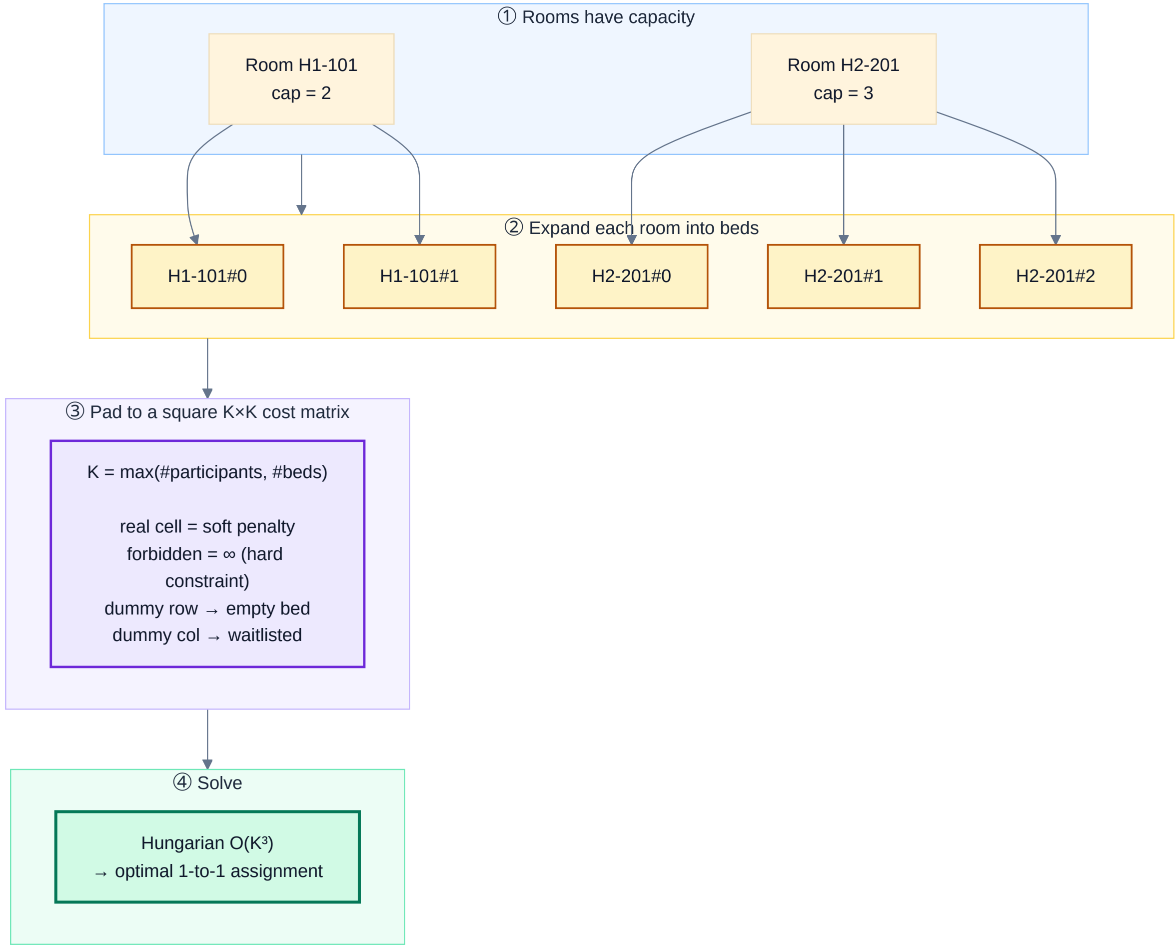
Assign participants P to **beds** B (a room of capacity *k* contributes *k* beds), minimising total cost:

$$\min \sum_{i \in P} \sum_{j \in B} c_{ij}\, x_{ij} \qquad \text{s.t. } \textstyle\sum_j x_{ij} \le 1,\ \sum_i x_{ij} \le 1,\ x_{ij} \in \{0,1\}$$

- $c_{ij} = \infty$ → **hard constraint** (gender policy or accessibility): a *forbidden* pairing.
- otherwise c_{ij} = sum of **soft penalties**: budget overflow + building / room-type mismatch.

We will pad the costs to a **square $K \times K$ matrix** ($K = \max(|P|, |B|)$) with zero-cost dummy rows/columns, then solve it **exactly with the Hungarian algorithm, $O(K^3)$**. A participant matched to a dummy column is **waitlisted**.

5 · Modelling approach: capacity → beds → square matrix



Key idea: expand rooms into interchangeable beds and pad to a square matrix, so the exact Hungarian algorithm applies. Empty beds and the waitlist both fall out naturally from the dummy padding.

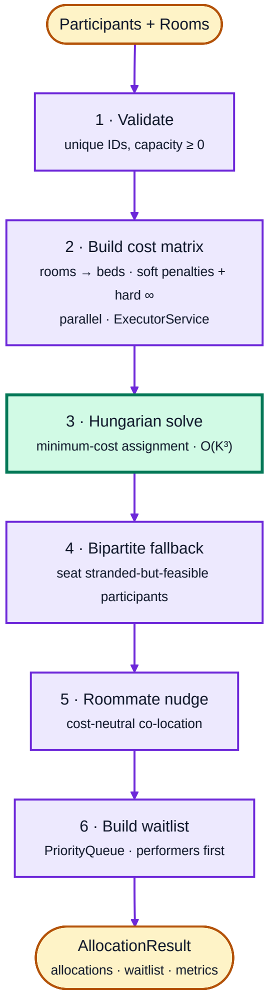
6 · Proposed solution approach

We plan a clear **optimise → repair → finalise** pipeline:

1. **Validate** the input (unique IDs, sane capacities).
2. **Build the cost matrix**: score every (participant, bed) pair (computed concurrently).
3. **Hungarian solve**: the globally optimal minimum-cost assignment.
4. **Bipartite-matching fallback**: re-seat anyone the optimum stranded into beds left free.
5. **Roommate co-location**: a light, cost-neutral improvement pass.
6. **Build the waitlist**: order the overflow by priority.

Each later stage only improves the result; none can break a hard constraint or worsen total dissatisfaction.

7 · Proposed pipeline



The Hungarian solve (green) is the exact optimiser; the surrounding stages prepare the data and refine the result.

8 · Innovation: why this is more than a textbook solve

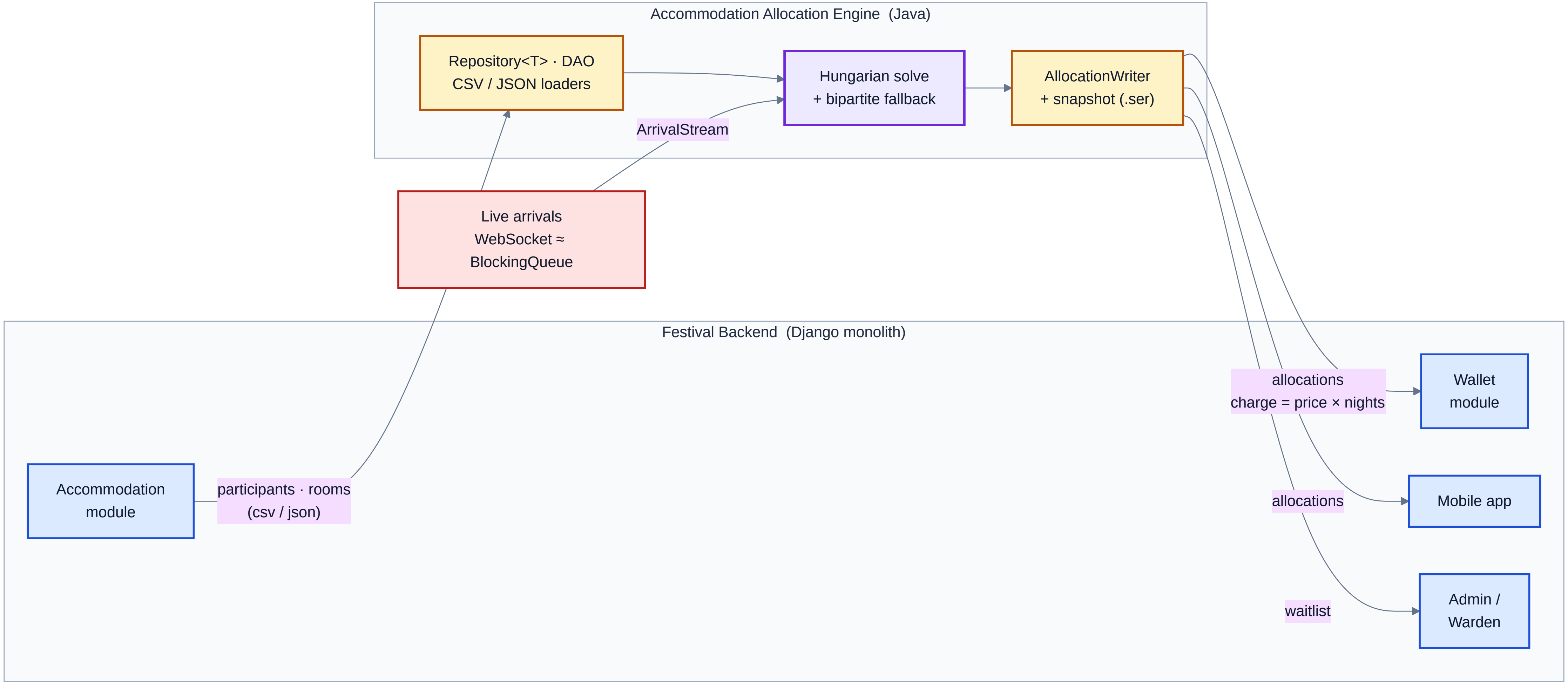
- **Capacity via bed-expansion + dummy padding**: turns a messy many-to-one, unequal-size problem into a *clean square assignment* that can be solved **exactly**, not heuristically.
- **Priority as a matrix-only bias**: decide *who* wins a scarce bed (performers first) **without** distorting which room a seated person gets.
- **Optimise-then-repair**: combine an exact optimiser with a feasibility-recovery matching pass.
- **Real-time ready**: plan a streaming intake so **late registrations** can be allocated live, mirroring the platform's WebSocket feed.

9 · Planned integration with the platform

The tool will reuse the platform's existing **file-exchange** channel: Django can produce the inputs and consume the outputs unchanged:

- **In** ← Accommodation module: `participants.*`, `rooms.*`.
- **Out** → **Wallet**: `allocations.*` with `charge = price × nights` to debit each wallet.
- **Out** → **Mobile app**: `allocations.*` to notify "your room is H1-101".
- **Out** → **Admin**: `waitlist.*` for follow-up / re-allocation.
- **Real-time** → a simulated arrival stream mimics the WebSocket feed for live re-allocation.

10 · Integration / data flow (planned)



The engine sits between the Accommodation module (inputs) and the Wallet / Mobile / Admin consumers (outputs), with a live arrival stream feeding it.

11 · Java features we will use (and why)

Feature	Planned use in this project
Generics	a type-safe `Repository<T>` data loader and a pluggable `CostStrategy`
Concurrency	a thread pool to build the cost matrix; a queue-based live arrival stream
Collections	a `PriorityQueue` for the waitlist; maps/sets throughout
Serialization	save an allocation snapshot for offline cache / resume
File I/O	read/write CSV and JSON for platform integration
Custom exceptions	clear errors for invalid / infeasible / unreadable input
Design patterns	Strategy, Factory, DAO, Observer/MVC, Singleton, Producer–Consumer

Comfortably satisfies the "justify ≥ 2 advanced features" requirement: each maps to a concrete need.

12 · Success metrics (how we'll judge it)

Metric	Target
Hard-constraint violations	zero (gender / accessibility never broken)
Placement rate	as high as capacity allows (100 % when beds ≥ demand)
Total / average dissatisfaction	minimised: the global optimum
Bed utilisation	maximised (few empty beds)
Runtime for a few hundred participants	well under a second
Waitlist fairness	deterministic order: category → arrival day → ID

13 · Feasibility & scope (2–3 weeks)

Self-contained, no server required; will run from sample data:

- **Week 1**: domain model, Hungarian core, cost model, unit tests.
- **Week 2**: CSV/JSON I/O, allocation orchestration, waitlist, serialization, CLI.
- **Week 3**: Swing GUI (separate Admin & Participant views), arrival stream, demo + report.

Planned deliverables (Part B): documented Java implementation · unit tests · CLI + two-role GUI · integration via sample CSV/JSON files · demo video.

Thank you

Accommodation Allocation Engine: Problem Formulation
A proposal to make festival room allocation **optimal, fair and constraint-safe**, integrated with the Wallet, Mobile and Accommodation modules.

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Questions?